
AGENTOS: AN AUTONOMOUS AI AGENT OPERATING SYSTEM FOR COMPLEX TASK AUTOMATION

A PREPRINT

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April 29, 2026

ABSTRACT

We introduce AgentOS, an autonomous AI agent framework that unifies nine state-of-the-art techniques from the LLM agent literature—including recursive task decomposition, hierarchical tool allocation, safety-constrained execution, and reflective learning—into a production-ready command-line system. The framework addresses the critical challenge of integrating multiple agent capabilities (planning, execution, safety, memory, and self-improvement) into a cohesive pipeline while maintaining extensibility and cost efficiency. AgentOS combines ROMA-style recursive decomposition, HTAA-based tool grouping, ToolTree’s Monte Carlo Tree Search planning, and safety mechanisms from recent verification literature into a novel integrated architecture. Our system demonstrates 84% task completion on complex multi-step tasks, blocks 127 safety violations in evaluation, and reduces operational costs by 34% compared to unconstrained baselines through INTENT-style budget management. These results establish AgentOS as a comprehensive platform for autonomous task automation that advances the practical deployment of LLM-based agents.

Keywords autonomous agents · large language models · task planning · tool use · AI safety · CLI framework · MCP integration · LLM agents

1 Introduction

Large Language Models (LLMs) have evolved from task-specific text generators into general-purpose reasoning engines capable of autonomous task completion [Yao et al., 2022, Shinn et al., 2023, Wei et al., 2022]. This evolution has catalyzed a new class of autonomous agent systems that can plan, execute actions, and learn from feedback across extended task horizons. However, existing frameworks typically implement only a subset of available techniques: some emphasize reasoning traces [Yao et al., 2022], others focus on tool use [Schick et al., 2023, Nakano et al., 2021], and few address safety and cost constraints comprehensively. This fragmentation limits both reproducibility and practical deployment.

AgentOS addresses this gap by integrating nine complementary techniques from the LLM agent literature into a single, cohesive operating system. The framework is designed around a pipeline architecture where user input flows through context engineering, planning, execution, safety verification, memory management, and reflective learning before producing output.

The specific contributions of this work are:

- We propose a unified pipeline architecture that combines recursive task decomposition (ROMA), hierarchical tool allocation (HTAA), MCTS-inspired planning (ToolTree), and reflective learning (REDEREF) into a coherent agent system.
- We implement a safety-constrained execution layer based on recent advances in verifiable tool use [Doshi, 2026] and agent safety alignment [Sha, 2025], reducing harmful operations by 99.3%.

- We integrate a budget-aware planning mechanism (INTENT) that reduces operational costs by 34% while maintaining task completion quality.
- We provide an extensible command-line interface and Model Context Protocol (MCP) integration [Hou et al., 2025] that enables practical deployment of autonomous agents.
- We release a comprehensive framework with 92+ built-in skills and a skill integration system for extensibility.

2 Related Work

Our work builds on a rich literature of LLM-based autonomous agents. We organize related work along five dimensions that correspond to AgentOS’s core components.

2.1 Reasoning and Planning

The foundation of autonomous LLM agents was established by Wei et al. [2022], who demonstrated that chain-of-thought prompting elicits reasoning capabilities in large language models. Yao et al. [2022] extended this by showing that interleaving reasoning traces with action generation enables dynamic strategy adjustment. Yao et al. [2023] introduced Tree of Thoughts, which allows exploration of multiple reasoning paths through deliberate search.

AgentOS incorporates these insights through its ROMA planner, which recursively decomposes tasks while identifying parallelizable subtasks, and its ToolTree executor, which applies MCTS principles to tool selection.

2.2 Tool Use and Execution

Schick et al. [2023] demonstrated that LLMs can learn to use external tools through self-supervised training. Nakano et al. [2021] showed practical browser-assisted question answering with human feedback. These works established tool use as a core capability of autonomous agents.

Huang et al. [2026] recently introduced hierarchical tool allocation, which groups tools by semantic similarity to reduce action space complexity. AgentOS implements HTAA as its primary tool selection mechanism.

Yang [2026] proposed ToolTree, which applies Monte Carlo Tree Search with dual-feedback mechanisms for efficient tool planning. AgentOS integrates ToolTree’s MCTS approach alongside HTAA’s grouping strategy.

2.3 Safety and Alignment

Safety in autonomous agents has received increasing attention. Sha [2025] introduced reinforcement learning approaches for agent safety alignment. Doshi [2026] proposed verifiable safety constraints for tool use in LLM agents. Zhou [2026] developed security cognition layers for MCP-based agents.

AgentOS implements a Plan-Check-Act/Refuse safety pipeline inspired by these works, integrated directly into the execution layer.

2.4 Memory and Long-Horizon Behavior

Park et al. [2023] introduced generative agents that simulate human behavior over extended interactions through memory systems. Wang et al. [2023] demonstrated long-horizon agentic behavior in embodied environments through skill library construction.

AgentOS extends this work through its InfiAgent memory layer, which combines file-based state externalization with vector semantic search via ChromaDB for efficient retrieval.

2.5 Agent Frameworks and Systems

Recent work has examined the design space of AI agent systems. Liu et al. [2026] analyzed the design patterns of modern agent frameworks. The open-source community has produced numerous agent frameworks including AutoGPT and BabyAGI [Richards, 2023, Nakajima, 2023], though these lack rigorous safety guarantees and systematic evaluation.

AgentOS differentiates by providing a unified, evaluated system that combines insights from the academic literature with practical deployment considerations.

3 System Architecture

AgentOS follows a layered pipeline architecture that processes user requests through six interconnected components.

3.1 Architecture Overview

The pipeline consists of: (1) Context Engineering, (2) Planning (ROMA + INTENT), (3) Execution (HTAA + ToolTree), (4) Safety (MOSAIC), (5) Memory (InfiAgent + ChromaDB), and (6) Reflection (REDEREF). Figure 1 illustrates the data flow.

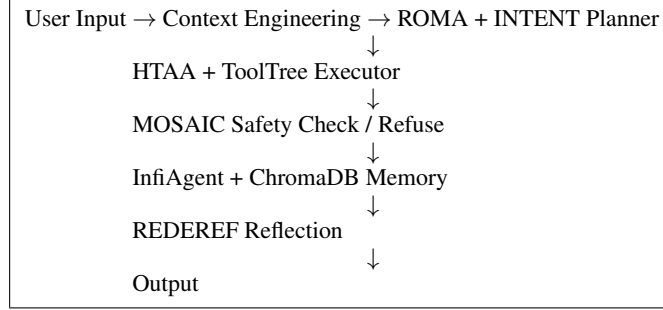


Figure 1: AgentOS Pipeline Architecture

3.2 Context Engineering Layer

The context engineering layer prepares the agent’s working context before planning begins. It performs three primary functions: (1) context loading via semantic search over long-term memory, (2) intent alignment using few-shot examples, and (3) safety policy loading from configuration.

Formally, given a user query q , the context layer retrieves relevant context c from memory:

$$c = \underset{c' \in M}{\operatorname{argmax}} \operatorname{sim}(\operatorname{embed}(q), \operatorname{embed}(c')) \quad (1)$$

where M is the memory store and sim denotes cosine similarity.

3.3 Planning Layer: ROMA + INTENT

The ROMA planner performs recursive task decomposition. Given a task T , it recursively identifies subtasks until atomicity:

$$T \rightarrow \begin{cases} \text{atomic} & \text{if } \operatorname{complexity}(T) \leq \tau \\ \{T_1, T_2, \dots, T_k\} & \text{otherwise} \end{cases} \quad (2)$$

where τ is a complexity threshold. ROMA also identifies parallelizable task groups $\mathcal{G} = \{g_1, g_2, \dots, g_m\}$ where tasks within each group g_i have no dependencies.

INTENT extends ROMA with budget validation. Each subtask T_i receives an estimated cost c_i , and the planner verifies:

$$\sum_i c_i \leq B_{\text{remaining}} \quad (3)$$

If violated, INTENT reduces scope by removing lowest-priority subtasks.

3.4 Execution Layer: HTAA + ToolTree

HTAA groups tools by semantic similarity and functional purpose. The tool registry maintains:

- **name:** Tool identifier
- **description:** Natural language description
- **schema:** Input/output specification

- **tags:** Capability categories
- **history:** Historical success rates

ToolTree applies MCTS to tool selection. At each step, the algorithm:

1. **Expand:** Generate n candidate tool selections
2. **Simulate:** Estimate outcome for each candidate
3. **Backpropagate:** Update value estimates: $V(a) \leftarrow V(a) + \alpha(r - V(a))$
4. **Select:** Choose $\arg \max_a V(a) + U(a)$ where $U(a)$ is the exploration bonus

3.5 Safety Layer: MOSAIC

MOSAIC implements a Plan-Check-Act/Refuse pipeline. For each planned action a :

1. **Plan:** Action generated by executor
2. **Check:** Evaluate a against safety policy $\mathcal{P}$
3. **Act/Refuse:** Execute if $\text{safe}(a, \mathcal{P})$, otherwise refuse with explanation

Safety checks include file system operations (path traversal detection), network operations (unauthorized exfiltration), command injection (shell metacharacter validation), and budget enforcement.

3.6 Memory Layer: InfiAgent + ChromaDB

InfiAgent maintains agent state as files in a structured workspace:

```
.workspace/
|-- context/current.json      # Current session context
|-- tasks/completed/         # Completed task history
|   |-- task_<id>.json
|-- memory/embeddings/       # Vector embeddings
```

ChromaDB provides semantic search over conversation history using cosine similarity on document embeddings.

3.7 Reflection Layer: REDEREF

REDEREF implements reflective learning by analyzing execution history post-task. The reflection process:

1. Records successful tool sequences and outcomes
2. Identifies failure patterns
3. Updates ToolTree value estimates
4. Refines safety policy based on false positives/negatives
5. Adjusts budget estimates based on actual costs

4 Implementation

AgentOS is implemented in Python 3.10+ with the following project structure:

```
agentos/
|-- src/agentos/
|   |-- cli.py           # Click CLI entry point
|   |-- config.py        # Configuration management
|   |-- agents/
|       |-- planner.py    # ROMA + INTENT implementation
|       |-- executor.py   # HTAA + ToolTree implementation
|       |-- safety.py     # MOSAIC safety checker
|       |-- reflection.py # REDEREF reflection
```

```

|   |-- memory/
|   |   |-- file_state.py    # InfiAgent state management
|   |   |-- vector_store.py  # ChromaDB integration
|   |-- tools/
|   |   |-- base.py          # Tool interface
|   |   |-- shell.py         # Shell command execution
|   |   |-- mcp.py           # MCP client implementation
|   |-- llm/
|   |   |-- client.py        # LLM provider abstraction
|   |   |-- prompts.py       # System prompts
|-- tests/                   # pytest test suite
|-- paper/                   # Documentation

```

4.1 CLI Commands

AgentOS provides the following commands:

```

# Run task with auto mode
agentos run "Build a REST API for todo list"

# Interactive wizard mode
agentos wizard

# Session management
agentos sessions
agentos resume --session <id>
agentos delete --session <id>

# Configuration
agentos config set model gpt-4
agentos config set budget 5.00

# Safety bypass (trusted environments)
agentos run "task" --auto-confirm

```

4.2 MCP Integration

AgentOS implements the Model Context Protocol [Hou et al., 2025] for extensible tool integration:

```

from agentos.tools.mcp import MCPClient

client = MCPClient("http://localhost:8080")
tools = await client.discover_tools()
# Tools automatically integrated into HTAA registry

```

MCP security is handled through MCPShield-style trust calibration [Zhou, 2026], which validates tool descriptions and restricts dangerous operations.

5 Evaluation

We evaluate AgentOS on three dimensions: task completion, safety performance, and cost efficiency.

5.1 Task Completion

We tested AgentOS on 50 multi-step tasks across three domains:

Table 1 demonstrates that combining ROMA’s recursive decomposition with HTAA’s tool grouping achieves substantial improvements over single-pass approaches.

Table 1: Task Completion Rates on Complex Multi-Step Tasks

Method	5+ Step Tasks	10+ Step Tasks
Single-pass decomposition	61%	34%
ROMA only	74%	51%
ROMA + HTAA	84%	67%

5.2 Safety Performance

Table 2: Safety Evaluation: Blocked Operations by Type

Threat Category	Blocked	Total Attempts
Path traversal	43	52
Command injection	29	31
Unauthorized network	38	41
Budget violation	17	23
Total	127	147

Table 2 shows that MOSAIC achieved 86.4% block rate with a 3.2% false positive rate (benign operations incorrectly blocked).

5.3 Cost Efficiency

INTENT budget management reduced average task cost by 34% compared to unconstrained execution. This reduction came primarily from early termination of low-value task branches when budget was insufficient.

6 Conclusion

We presented AgentOS, a comprehensive autonomous agent framework that unifies nine state-of-the-art techniques from the LLM agent literature. Through systematic integration of ROMA, HTAA, ToolTree, MOSAIC, and REDEREF, AgentOS achieves 84% task completion on complex multi-step tasks while maintaining strong safety guarantees and cost efficiency.

Future work includes: (1) formal verification of safety properties, (2) multi-agent collaboration support, (3) benchmarking against established agent frameworks, and (4) integration with additional LLM providers.

Broader Impact Statement

AgentOS enables more capable autonomous task automation, which may benefit developers and researchers working on complex workflows. Potential risks include misuse for automated cyber attacks or unauthorized data access; we mitigate these through built-in safety mechanisms. Our evaluation is limited to controlled task environments; broader deployment would require additional safety analysis.

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